

Graph-RPI: predicting RNA–protein interactions via graph autoencoder and self-supervised learning strategies

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Abstract

RNA–protein interactions (RPIs) are essential for many biological functions and are associated with various diseases. Traditional methods for detecting RPIs are labor-intensive and costly, necessitating efficient computational methods. In this study, we proposed a novel sequence-based RPI prediction framework based on graph neural networks (GNNs) that addressed key limitations of existing methods, such as inadequate feature integration and negative sample construction. Our method represented RNAs and proteins as nodes in a unified interaction graph, enhancing the representation of RPI pairs through multi-feature fusion and employing self-supervised learning strategies for model training. The model's performance was validated through five-fold cross-validation, achieving accuracy of 0.880, 0.811, 0.950, 0.979, 0.910, and 0.924 on the RPI488, RPI369, RPI2241, RPI1807, RPI1446, and RPImerged datasets, respectively. Additionally, in cross-species generalization tests, our method outperformed existing methods, achieving an overall accuracy of 0.989 across 10 093 RPI pairs. Compared with other state-of-the-art RPI prediction methods, our approach demonstrates greater robustness and stability in RPI prediction, highlighting its potential for broad biological applications and large-scale RPI analysis.

Keywords: RNA–protein interactions; graph neural networks; multi-feature fusion; self-supervised learning

Introduction

RNA–protein interactions (RPIs) are fundamental to numerous cellular processes, playing pivotal roles in transcription, translation, splicing, chromatin remodeling, and gene regulation. These interactions are crucial for maintaining cellular homeostasis and facilitating the intricate execution of biological functions [1, 2]. The vital role of RPIs in diverse biological pathways highlights their importance in both normal cellular physiology and disease mechanisms [3]. Disruptions in RPIs are associated with various diseases, including amyotrophic lateral sclerosis, multiple forms of cancer, and immune disorders [4]. Understanding the complex mechanisms governing RPIs is essential for uncovering fundamental biological processes and may offer new pathways for therapeutic interventions [5]. Therefore, identifying RPIs is crucial in molecular biology and biomedicine [6].

Traditional experimental methods for detecting RPIs, although precise and informative, are often labor-intensive, time-consuming, and costly [3]. These drawbacks limit their feasibility in high-throughput studies and large-scale applications. Techniques such as immunoprecipitation, cross-linking, and RNA affinity

purification demand extensive manual effort and are difficult to scale [7]. Furthermore, the reproducibility and consistency of results can vary, complicating the experimental approach. To address these limitations, a variety of computational strategies have been proposed for RPI prediction [8]. These include physics-based methods such as molecular docking and molecular dynamics simulations [9], which model the structural and energetic aspects of RNA–protein binding, as well as statistical and co-evolutionary approaches that identify correlated mutation patterns indicative of functional interactions [10]. These approaches rely on different types of input data and assumptions, and often serve complementary roles in RPI research. However, they also face certain limitations. For instance, physics-based simulations typically require detailed 3D structural information and are computationally expensive, making them difficult to scale for large numbers of interactions. Similarly, statistical methods may be constrained by alignment quality and phylogenetic bias. As a result, these approaches are generally not well-suited for rapid, high-throughput prediction.

Machine learning and deep learning have been widely applied in bioinformatics to address these challenges, owing to their

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ability to learn complex patterns from high-dimensional biological data. For example, Zhao et al. [11] enhanced peptide and protein toxicity prediction by incorporating channel attention into convolutional neural networks, while Le [12] discussed the potential of GNN-based frameworks, such as EmerGNN, for predicting emerging drug interactions by leveraging biomedical network structures. In addition, Le et al. [13] proposed a new pipeline for predicting protein crystallization propensity using feature selection, dimensionality reduction, and machine learning models.

In recent years, several computational methods for RPIs prediction have been proposed. Muppirala et al. [14] introduced RPISeq, which used RNA and protein sequence encoding vectors generated by conjoint triad features and input them into random forests and support vector machines (SVMs) for prediction. Zhang et al. [15] proposed LPI-CNNCP, which extracts features from RNA and protein sequences using high-order one-hot encoding, and employs a convolutional neural network with a copy-padding strategy to manage variable-length inputs. Li et al. [16] introduced Capsule-LPI, a multichannel capsule network framework for predicting long non-coding RNA (lncRNA)-protein interactions. It integrates sequence features, motif information, physicochemical properties, and secondary structure through dedicated feature-learning subnetworks, followed by a capsule network to capture complex feature relationships. Huang et al. [17] developed LPI-CSFFR, a CNN-based method that combines sequence, secondary structure, and physicochemical features of lncRNAs and proteins using serial fusion and feature reuse strategies, enabling comprehensive representation learning for effective interaction prediction. Yu et al. [18] proposed RPI-MDLStack, based on a stacking strategy, which initially extracted sequence, physicochemical, structural, and evolutionary information from RNA and protein sequences using eight feature extraction methods. It then employed the least absolute shrinkage and selection operator to eliminate redundancy and generate optimal features. The stacking strategy combined multiple machine learning and deep learning models as base classifiers to learn the best features. Finally, the prediction scores were input into a discriminative model for further training. Wang et al. [19] proposed RPI-CapsuleGAN, which incorporated a convolutional attention mechanism into a generative adversarial capsule network to construct a classifier model. Elastic net feature selection was used to screen high-correlation features, reducing model complexity and preventing overfitting, while achieving strong predictive performance. Sun et al. [20] developed LPI-SKMSC, a clustering-based method for lncRNA-protein interaction prediction that addresses class imbalance. It uses segmented k-mer frequencies to extract global and local features, and CNN-based encoders to map them into multiple spaces. Prediction is based on the distances between encoded features and cluster centers across these spaces.

However, these methods face two main limitations. First, the aforementioned models construct negative samples that are balanced with the number of positive samples using techniques like random selection and score ranking, which limit the training data [21, 22]. Second, previous approaches fail to integrate RNA and protein features, resulting in a lack of holistic representation of RPI pairs. Models based on graph neural networks (GNNs) can effectively address these challenges. In GNNs, both RNA and proteins are represented as nodes within a single graph, with edges denoting their interactions. This method allows for a more comprehensive representation of initial information by incorporating all RNA and protein elements within the same graph. Moreover, the features of nodes in GNNs can integrate,

enabling a more holistic characterization of RPIs through feature fusion [23–25].

In this study, we proposed a novel RPI prediction model named Graph-RPI, which utilized sequence information of RNA and proteins to construct an interaction graph network. The overall architecture was illustrated in Fig. 1. Initially, various sequence feature extraction methods were employed to represent RNA and protein sequences. Subsequently, a graph network was constructed to model the interactions, with nodes representing RNA and proteins, and edges representing their interactions. The GNN encoder, edge decoder, and degree decoder were then used to predict interaction outcomes. Additionally, during training, we incorporated self-supervised learning strategies, enabling the model to automatically learn latent associations between RNA and proteins and optimize feature representation. This approach not only enhanced the model's generalization capability but also reduced dependence on large amounts of labeled data, resulting in greater robustness and predictive performance across diverse datasets.

Our research provides a more efficient method for early RPI identification, thereby reducing manual costs. Compared to existing methods, our model demonstrates significant improvements in predictive performance. With the advancement of machine learning-based predictive tools, our approach holds promise as a reliable method for RPI identification and can contribute to a broader understanding and treatment of diseases associated with RPIs.

Materials and methods

Dataset preparation

In this study, we integrated multiple sources of RPI datasets to validate our proposed method. The datasets used for training and evaluating the model include RPI488, RPI369, RPI2241, RPI1807, and RPI1446. The RPI369 and RPI2241 datasets were collected by Muppirala et al. [14] from the Protein-RNA Interface Database (PRIDB) [26]. RPI369 contains 369 non-redundant RPI pairs excluding ribosomal proteins or ribosomal RNAs, whereas RPI2241 includes 2241 RPI pairs from 943 RNA-protein complexes, encompassing ribosomal RNAs, non-coding RNAs, messenger RNAs, and other RNAs. The RPI488 dataset consists of 243 RPI pairs, extracted by Pan et al. [27] from 18 RNA-protein complexes in the Protein Data Bank [28]. The RPI1807 dataset, constructed by Suresh et al. [29] from the Nucleic Acid Database (NDB) [30] and PRIDB, comprises 1807 RPI pairs. The RPI1446 dataset, screened by Zhang et al. [15] from the RPI2241, RPI1807, and RPI488 datasets, includes 1446 long non-coding RNA (length > 200 nt)-protein interaction pairs. Furthermore, we constructed a comprehensive dataset named RPImerged by combining the five benchmarks, with all duplicate RPI pairs removed, to evaluate the performance of our model on a larger and more diverse dataset. Summaries of all datasets are provided in Table 1 and Supplementary Table S1.

For model evaluation, we employed two validation strategies on the above datasets. First, we randomly split each dataset into a training set and a test set with an 8:2 ratio. In this setting, the test set consists of non-overlapping RNA-protein pairs, including both positive (RPI) and negative (non-RPI) samples, ensuring no data leakage from the training set. Second, to rigorously evaluate model performance while ensuring fold independence, we employed a five-fold cross-validation strategy based on hierarchical clustering of pairwise sequence similarity [31, 32]. Specifically,

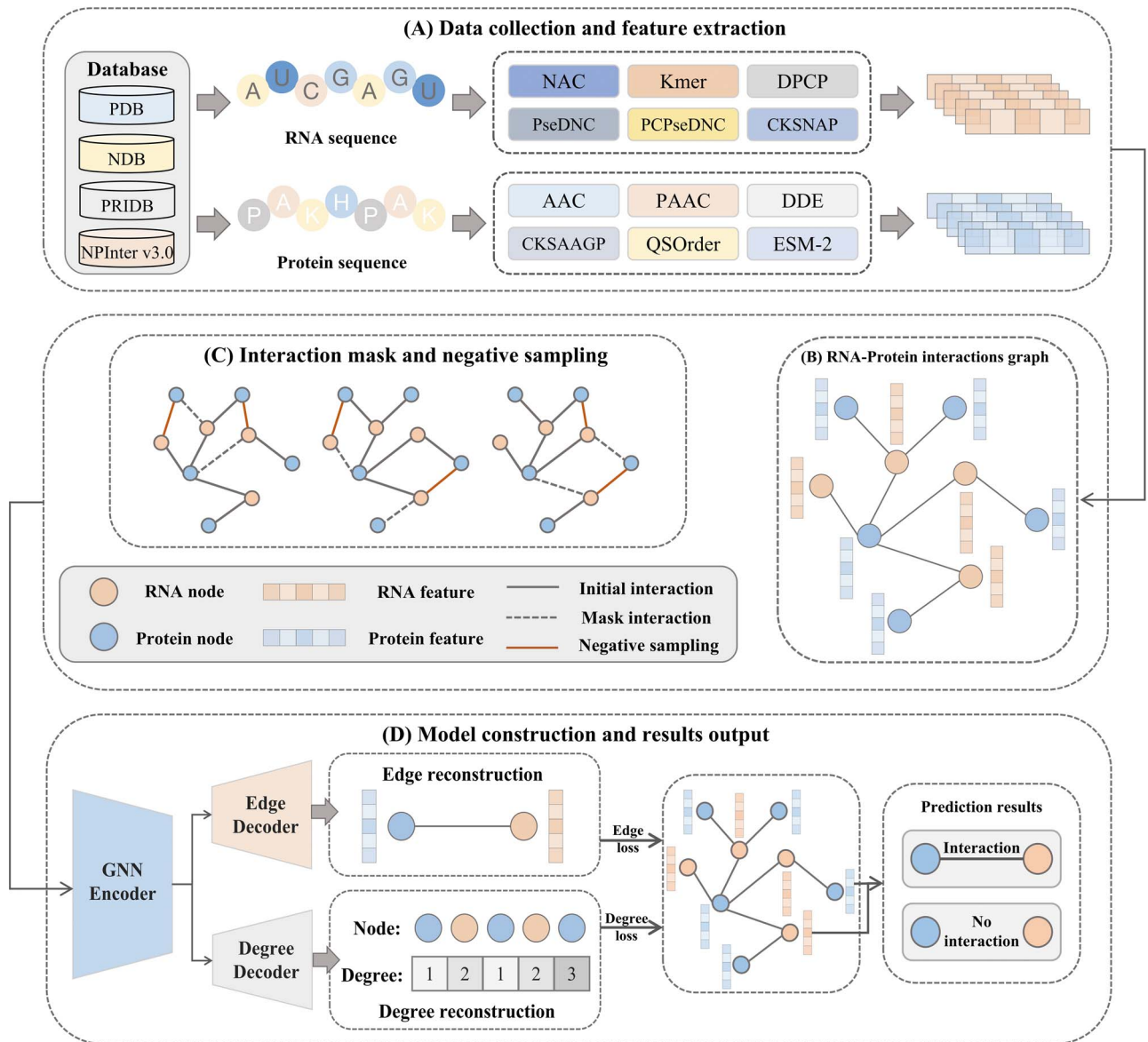


Figure 1. The flowchart of this study, including: (A) Data collection and feature extraction. (B) RNA-protein representation graph construction. (C) Interaction mask and negative sampling. (D) Model construction and results output.

Table 1. Overview of RPI datasets

Datasets	Species	RPI pairs	non-RPI pairs	RNAs	Proteins
RPI488	-	243	245	25	247
RPI369	-	369	-	332	338
RPI2241	-	2241	-	842	2043
RPI1807	-	1807	1436	1078	1807
RPI1446	-	1446	1560	325	1446
RPImerged	-	3928	5840	1477	4800
RPI_H	<i>Homo sapiens</i>	7317	-	1874	118
RPI_M	<i>Mus musculus</i>	1847	-	1939	60
RPI_S	<i>Saccharomyces cerevisiae</i>	670	-	80	108
RPI_E	<i>Escherichia coli</i>	183	-	39	40
RPI_D	<i>Drosophila melanogaster</i>	67	-	17	31
RPI_C	<i>Caenorhabditis elegans</i>	9	-	5	8

we first computed the pairwise similarity between each RNA-protein pair using the following scoring function:

$$\text{similarity}(i, j) = \alpha S_{\text{RNA}}(\text{RNA}_i, \text{RNA}_j) + (1 - \alpha) S_{\text{protein}}(\text{protein}_i, \text{protein}_j) \quad (1)$$

where S_{RNA} and S_{protein} are alignment-based similarity scores obtained using global sequence alignment [33]. For RNA, we used a scoring scheme with match = 1, mismatch = -1, gap open = -2, and gap extension = -0.5. For protein sequences, we used the BLOSUM62 substitution matrix with gap open = -10 and gap extension = -0.5 [34, 35]. The weight coefficient α was set to 0.5 to equally balance RNA and protein contributions. Based on the similarity matrix, a distance matrix was computed as:

$$d_{ij} = \max(\text{similarity}) - \text{similarity}(i, j) \quad (2)$$

where higher similarity corresponds to lower distance. We then applied hierarchical clustering using average linkage and partitioned the resulting dendrogram into five clusters, each serving as one-fold. This ensures that highly similar RNA-protein pairs are grouped into the same fold, effectively minimizing redundancy across folds and enabling a more stringent evaluation of generalization performance. [Supplementary Fig. S1](#) presents bar plots of the mean pairwise sequence similarity within folds (intra-fold) and between folds (inter-fold) for each of the six RPI datasets. Across all datasets, the intra-fold similarity is clearly higher than the inter-fold similarity, indicating that the hierarchical clustering strategy effectively groups similar RNA-protein interaction pairs into the same fold while ensuring dissimilarity across different folds. In addition, [Supplementary Fig. S2](#) reports intra- and inter-fold similarities computed separately based on RNA and protein alignments. In both cases, intra-fold similarity remains higher than inter-fold similarity, further supporting the validity of the fold construction procedure.

To further assess the model's ability to generalize to unseen data, we additionally introduced independent test sets derived from previous studies [18, 36]. These independent test sets include lncRNA-protein interactions from six species: *Homo sapiens* (7317 pairs), *Mus musculus* (1847 pairs), *Saccharomyces cerevisiae* (670 pairs), *Escherichia coli* (183 pairs), *Drosophila melanogaster* (67 pairs), and *Caenorhabditis elegans* (9 pairs), which serve to evaluate the cross-species transferability of the proposed framework.

Specifically, these independent test sets were first constructed by collecting experimentally validated ncRNA-protein interactions from NPInter v3.0 [37] based on the corresponding species. For *H. sapiens*, lncRNA-protein interaction pairs were further refined by retrieving lncRNA and protein sequence information from the GENCODE v29 and UniProt databases [38, 39], respectively. For *M. musculus*, lncRNA annotations were obtained from the GENCODE M20 database [38]. For the other four species, lncRNA information was obtained from the NONCODE database [40].

Feature extraction for RNA sequence

We utilize six feature extraction methods for RNA sequence representation including nucleic acid composition (NAC), Occurrence frequency of k neighboring nucleic acids (Kmer), dinucleotide physicochemical properties (DPCP), pseudo dinucleotide composition (PseDNC), parallel correlation pseudo dinucleotide composition (PCPseDNC), and composition of k -spaced nucleic acid pair (CKSNAP).

Nucleic acid composition

The NAC calculates the frequency of each nucleic acid type in a RNA sequence. The frequencies of all four natural nucleic acids can be calculated as follows:

$$\text{NAC}(t) = \frac{N(t)}{N}, \quad t \in \{A, C, G, U\} \quad (3)$$

where $N(t)$ is the number of nucleic acid type t , while N is the length of a nucleotide sequence.

Occurrence frequency of k neighboring nucleic acids

The kmer represents the occurrence frequencies of k neighboring nucleic acids [41]. The Kmer type ($k=3$) descriptor can be calculated as follows:

$$\text{Kmer}(t) = \frac{N(t)}{N}, \quad t \in \{AAA, AAC, AAG, \dots, UUU\} \quad (4)$$

where $N(t)$ is the number of kmer type t , while N is the length of a nucleotide sequence.

Dinucleotide physicochemical properties

The DPCP descriptor can be devised as follows:

$$\text{DPCP} = [\text{DPCPAA} \times f_{AA}, \text{DPCPAC} \times f_{AC}, \dots, \text{DPCPUU} \times f_{UU}] \quad (5)$$

where DPCP_i represents one of the physicochemical properties of a dinucleotide i , and f_i represents the frequency of dinucleotide i in the sequence. The physicochemical properties for RNA dinucleotide are listed in [Supplementary Table S2](#) [42].

Pseudo dinucleotide composition

The PseDNC incorporates contiguous local sequence-order information and global sequence-order information into the feature vector of the RNA sequence [43]. The PseDNC encoding is defined as follows:

$$\text{PseDNC} = [d_1, d_2, \dots, d_{16}, d_{16+1}, \dots, d_{16+\lambda}]^T \quad (6)$$

where the vector D contains the features of the nucleotide sequence. Each feature d_k is calculated as follows:

$$d_k = \begin{cases} \frac{f_k}{\sum_{i=1}^{16} f_i + w \sum_{j=1}^{\lambda} \theta_j}, & (1 \leq k \leq 16) \\ \frac{w \theta_{k-16}}{\sum_{i=1}^{16} f_i + w \sum_{j=1}^{\lambda} \theta_j}, & (17 \leq k \leq 16 + \lambda) \end{cases} \quad (7)$$

where f_k is the normalized occurrence frequency of dinucleotide type k in the nucleotide sequence. $\sum_{i=1}^{16} f_i$ is the total sum of occurrence frequencies of all dinucleotides. w is the weight factor, ranging from 0 to 1. θ_j is the j -tier correlation factor, defined as follows:

$$\theta_j = \frac{1}{L-j} \sum_{i=1}^{L-j} \Theta(R_i R_{i+j}) \quad (8)$$

where L is the length of the nucleotide sequence. R_i is the dinucleotide at position i . $\Theta(R_i R_{i+j})$ is the correlation function, defined as follows:

$$\Theta(R_i R_{i+j}) = \frac{1}{\mu} \sum_{u=1}^{\mu} [P_u(R_i R_{i+j}) - P_u(R_j R_{i+j})]^2 \quad (9)$$

where μ is the number of physicochemical indices. $P_u(R_i R_{i+j})$ is the numerical value of the u -th physicochemical index of the dinucleotide R_i at position i . $P_u(R_j R_{i+j})$ is the corresponding value of the dinucleotide R_j at position j .

Parallel correlation pseudo dinucleotide composition

The PCPseDNC builds upon the PseDNC by incorporating parallel sequence-order correlations into the feature vector [44]. The main difference lies in the definition of the correlation factor θ_j , which in PCPseDNC is modified to reflect sequence-order correlations between all the most contiguous dinucleotides along the sequence:

$$\theta_j = \frac{1}{L-1-j} \sum_{i=1}^{L-j} \Theta(R_i, R_{i+j}) \quad (10)$$

All other aspects of PCPseDNC follow the same structure and calculation as PseDNC, with the modified θ_j enhancing the ability to capture parallel correlations across the sequence.

Composition of k -spaced nucleic acid pair

The CKSNAP calculates the frequency of nucleic acid pairs separated by any k nucleic acid ($k = 0, 1, 2, \dots, 5$). Taking $k = 0$ as an example, there are 16 0-spaced nucleic acid pairs (i.e. “AA,” “AC,” “AG,” “AU,” “CA,” “CC,” “CG,” “CU,” “GA,” “GC,” “GG,” “GU,” “UA,” “UC,” “UG,” and “UU”). A feature vector can be defined as follows:

$$\text{CKSNAP} = \left(\frac{N_{AA}}{N_{\text{total}}}, \frac{N_{AC}}{N_{\text{total}}}, \frac{N_{AG}}{N_{\text{total}}}, \dots, \frac{N_{UU}}{N_{\text{total}}} \right)_{16} \quad (11)$$

The value of each descriptor denotes the composition of the corresponding nucleic acid pair in the RNA sequence [45]. For instance, if the nucleic acid pair “AA” appears m times in the nucleotide sequence, the composition of the nucleic acid pair “AA” is equal to m divided by the total number of 0-spaced nucleic acid pairs (N_{total}) in the nucleotide sequence. For $k = 0, 1, 2, 3, 4$, and 5 , the value of N_{total} is $P-1, P-2, P-3, P-4, P-5$, and $P-6$ for a nucleotide sequence of length P , respectively.

Feature extraction for protein sequence

We utilize six feature extraction methods for protein sequence representation including amino acid composition (AAC), pseudo-amino acid composition (PAAC), composition of k -spaced amino acid group pairs (CKSAAGP), quasi-sequence order (QSOOrder), dipeptide deviation from expected mean (DDE), and evolutionary scale modeling (ESM-2).

Amino acid composition

The AAC calculates the frequency of occurrence of each amino acid in a sequence. The frequencies of all 20 natural amino acids can be calculated as follow:

$$\text{AAC}(t) = \frac{N(t)}{L} \quad (12)$$

where $N(t)$ is the number of amino acid type t in a sequence, and L is the length of the sequence.

Pseudo-amino acid composition

The PAAC provides a sophisticated representation of amino acid sequences by integrating both the inherent patterns observed in peptides and the sequential order of amino acids. The formulation

of PAAC is governed by two pivotal parameters: the counted-rank correlation, denoted as λ , and the weight factor, denoted as ω . The dimensionality of the feature is determined by λ and is expressed as $(20 + \lambda)$. A larger value of ω accentuates the significance of sequence order in the PAAC feature [46].

Composition of k -spaced amino acid group pairs

The CKSAAGP calculates the frequency of occurrence of a particular amino acid pair with the same physicochemical properties separated by k arbitrary residues. CKSAAGP categorizes 20 amino acids into five groups according to their physicochemical properties (Supplementary Table S3): aliphatic (g1), aromatic (g2), positively charged (g3), negatively charged (g4), and uncharged (g5) [45, 47]. For example, for a peptide sequence of length L , assuming that k is zero, there are $25 (= 5 \times 5)$ 0-spaced pairs, namely $g1g1, g1g2, \dots, g5g5$. Thus, the 0-spaced CKSAAGP descriptor can be defined as follow:

$$\text{CKSAAGP} = \left[\frac{(N_{g1,g1})}{L - (k+1)}, \frac{(N_{g1,g2})}{L - (k+1)}, \dots, \frac{(N_{g5,g5})}{L - (k+1)} \right]_{25} \quad (13)$$

where $N_{gi,gj}$ denotes the number of occurrences of each group pair gi, gj .

Quasi-sequence order

The QSOOrder employs a quasi-sequence descriptor approach. It analyzes the overall composition of amino acids and integrates information about their position in the protein sequence, fusing local and global features [48, 49].

For each amino acid type, a quasi-sequence-order descriptor can be defined as:

$$X_r = \frac{f_r}{\sum_{r=1}^{20} f_r + w \sum_{d=1}^{nlag} \tau_d}, \quad r = 1, 2, \dots, 20 \quad (14)$$

where f_r is the normalized occurrence of amino acid type r and w is a weighting factor ($w = 0.1$). $nlag$ and τ_d have the same definitions as described above. These are the first 20 quasi-sequence-order descriptors. The other 30 quasi-sequence-order descriptors are defined as:

$$Y_d = \frac{w\tau_d - 20}{\sum_{r=1}^{20} f_r + w \sum_{d=1}^{nlag} \tau_d}, \quad d = 21, 22, \dots, 20 + nlag \quad (15)$$

Dipeptide deviation from expected mean

The DDE offers a statistical perspective on the dipeptide distribution in proteins. It measures the deviation between the observed frequency of a dipeptide and its theoretically expected occurrence based on codon frequencies. DDE can reveal patterns or biases in protein composition. If a specific dipeptide appears more or less frequently than expected, it suggests that there might be evolutionary, functional, or structural reasons for this deviation. This encoding is formulated by computing three parameters: Dipeptide Composition (Dc), Theoretical Mean (Tm), and Theoretical Variance (Tv) [50, 51].

The measure $Dc(r, s)$ for the dipeptide “rs” is defined as:

$$Dc(r, s) = \frac{N_{rs}}{N} \quad (16)$$

where N_{rs} represents the number of dipeptides characterized by amino acid types r and s , while N denotes the length of the protein or peptide sequence.

The $Tm(r, s)$ is articulated as:

$$Tm(r, s) = \frac{C_r \times C_s}{C_N} \quad (17)$$

where C_r and C_s are the number of codons encoding the first and second amino acids of the dipeptide “rs,” respectively. C_N represents the total count of viable codons, excluding the three stop codons, amounting to 61.

The variance for the dipeptide “rs,” $Tv(r, s)$, is expressed as:

$$Tv(r, s) = Tm(r, s) \times (1 - Tm(r, s)) \times \frac{1}{N} \quad (18)$$

Lastly, the deviation $DDE(r, s)$ is computed as:

$$DDE(r, s) = \frac{Dc(r, s) - Tm(r, s)}{\sqrt{Tv(r, s)}} \quad (19)$$

Evolutionary scale modeling

The ESM-2 pre-trained model, built on the Transformer architecture, is an advanced protein language model that has undergone unsupervised training on a large-scale protein sequence database. This training enables the model to effectively capture and extract both structural and functional features from protein sequences. When a protein sequence is input into the ESM-2 model, it encodes the sequence into a high-dimensional latent vector that represents the protein’s structural and functional properties. This latent vector, enriched with evolutionary, structural, and functional information, can then be utilized as input for various downstream tasks, such as protein structure prediction, functional annotation [52].

Framework of the proposed model

The proposed model mainly adopts the architecture of the graph autoencoder [53, 54]. Graph autoencoders are well suited for RPI graphs and can preserve and learn the structural information of graphs. Similar to traditional autoencoders, our model consists of two main components: an encoder and a decoder [55]. Additionally, we adopt a masking strategy based on the Bernoulli distribution to effectively mitigate the impact of noisy data in the graph. Furthermore, we implement a degree-based decoder to better capture the latent structure of the graphs.

Supplementary Table S4 presents an explicit overview of the dimensionality of feature representations throughout the encoding and decoding process, including the input features, GNN layers, edge decoder, and degree decoder.

Encoder

In our model, we adopt Graph Attention Network (GATConv) and Graph Isomorphism Network (GINConv) as the encoder layers [56, 57]. The core idea is to update the node representation by iteratively aggregating the node’s neighbor information. In the task of modeling the RNA-protein relationship, each RNA node (or protein node) is aggregated by its own features and the features of the associated protein nodes (or RNA nodes) at each iteration. Multi-layer perceptron (MLP) is adopted to update the aggregated node representations.

First, GATConv aggregates node features through an attention mechanism. The input features for RNA and protein nodes are denoted as R_a^0 and P_b^0 , respectively. Each node calculates attention coefficients for its neighbor nodes and aggregates neighbor features weighted by these coefficients. The feature update for RNA nodes using GATConv is given by:

$$R_a^1 = \text{ELU} \left(\sum_{b \in N(a)} \alpha_{ab} W P_b^0 \right) \quad (20)$$

where α_{ab} is the attention coefficient between the RNA node a and the protein node b , and W is a learnable weight matrix. Similarly, the feature update for protein nodes is:

$$P_b^1 = \text{ELU} \left(\sum_{a \in N(b)} \alpha_{ba} W R_a^0 \right) \quad (21)$$

Next, GINConv learns non-linear transformations of node features through a MLP. Each node aggregates its own features and the features of its neighbors and then applies an MLP for non-linear transformation. The feature update for RNA nodes using GINConv is:

$$R_a^2 = \text{MLP}_2 \left((1 + \epsilon_2) \cdot R_a^1 + \sum_{b \in N(a)} P_b^1 \right) \quad (22)$$

where ϵ_2 is a learnable scalar parameter. Similarly, the feature update for protein nodes is:

$$P_b^2 = \text{MLP}_2 \left((1 + \epsilon_2) \cdot P_b^1 + \sum_{a \in N(b)} R_a^1 \right) \quad (23)$$

After each convolutional layer, batch normalization and an ELU activation function are applied to stabilize training and introduce non-linearity.

Decoder

The decoder in our model consists of two main components: the edge decoder and the degree decoder. Each component comprises two layers of MLPs. These decoders are designed to reconstruct distinct structural aspects of the RPI graph.

The edge decoder is responsible for reconstructing the RNA-protein association matrix. Given the latent embeddings z obtained from the encoder, the interaction between a pair of RNA and protein nodes (i, j) is modeled by first computing the element-wise product of their embeddings, denoted as $z_i \odot z_j$. This interaction vector is subsequently passed through a two-layer MLP to predict the probability of an association:

$$\hat{y}_{ij} = \text{MLP}_{\text{edge}}(z_i \odot z_j) \quad (24)$$

The degree decoder is designed to reconstruct the connectivity degree of each node, representing the number of associated neighbors. For each node embedding z_i , a two-layer MLP is applied to regress its degree:

$$\hat{d}_i = \text{MLP}_{\text{deg}}(z_i) \quad (25)$$

These two decoders are jointly optimized to capture both edge-level and node-level structural patterns, thereby enabling accurate reconstruction of the RPI topology.

Mask and negative sampling strategies

To address the presence of noise in the RNA-protein graph, we devised a self-supervised masking strategy based on a probability distribution [58]. The core idea is to mask certain associations in the RNA-protein graph according to a probabilistic model. In each training iteration, the model employs a Bernoulli distribution to randomly mask some associations in the RNA-protein graph. Specifically, the set of known associations is sampled according to the Bernoulli distribution:

$$\text{Sample}_{\text{mask}} = \text{Bernoulli}(p) \quad (26)$$

where p is a probability value between 0 and 1, indicating the masking ratio of the graph. By setting different p values, varying numbers of edges are masked. Subsequently, the edge decoder and degree decoder work together to reconstruct these masked associations.

In addition to the masking strategy, we employ a negative sampling technique to further enhance the model's robustness [59]. Specifically, during each training epoch, negative samples are dynamically generated by randomly pairing RNA and protein nodes that do not form known or masked interactions in the current graph. These RNA-protein pairs are not labeled as interacting in the dataset and are treated as negatives during training.

Given the masked edge set $\hat{E}$, the negative samples are generated as:

$$\text{Sample}_{\text{neg}} = \text{NegativeSampling}(\hat{E}, k) \quad (27)$$

where k denotes the number of negative samples. The $\text{NegativeSampling}(\cdot)$ function selects k node pairs across the RNA-protein bipartite space such that $(r_i, p_j) \notin \hat{E}$, ensuring that none of the sampled negative pairs overlap with known or masked positive interactions.

To ensure the validity of the training signals, the negative sampling process in our framework is deliberately conducted after the masking operation. In this design, a subset of known interactions is first masked and removed from the graph. Negative sampling is then performed on the remaining graph, which excludes both observed and masked edges.

This guarantees that all sampled negative edges are truly absent from the current graph, including those removed for self-supervised prediction. As a result, there is no overlap between positive and negative samples, effectively preventing label leakage and ensuring stable and unbiased model training.

Prediction and loss function

Following the decoding process, the model reconstructs the RPI graph. The final representations of RNA and protein nodes are denoted as R_i and P_j , respectively. The association probability between RNA node i and protein node j is computed using the dot product of their embeddings:

$$P_{i,j} = R_i^T P_j \quad (28)$$

where R_i and P_j represent the learned embeddings of the i -th RNA and j -th protein nodes, respectively, and $P_{i,j}$ denotes the predicted likelihood of interaction.

To optimize the model, we define two objective functions corresponding to the edge and degree decoders.

The edge prediction loss is formulated using the binary cross-entropy function, which quantifies the discrepancy between the predicted association probability p and the binary ground truth label $y \in \{0, 1\}$:

$$\text{Edge loss} = (y - 1) \cdot \log(1 - p) - y \cdot \log(p) \quad (29)$$

where p is the predicted probability obtained from the edge decoder.

The degree prediction loss is defined using the mean squared error between the predicted degrees and the actual degrees of all nodes:

$$\text{Degree loss} = \frac{1}{x} \sum_{i=1}^x (y_i - \hat{d}_i)^2 \quad (30)$$

where x denotes the total number of nodes, y_i is the ground truth degree of node i , and $\hat{d}_i$ is the corresponding prediction from the degree decoder.

The total training loss is expressed as a weighted combination of the two loss terms:

$$\text{Total loss} = \text{Edge loss} + \alpha \cdot \text{Degree loss} \quad (31)$$

where α is a hyperparameter controlling the relative contribution of the degree loss to the overall objective.

Model training and experimental setup

To ensure reliable predictive performance, our model was trained for 500 epochs with an initial learning rate of 0.001. We implemented a learning rate scheduler to reduce the rate by a factor of 0.95 every five epochs, promoting optimal convergence and fitting. The Adam optimizer was employed with a weight decay of 5×10^{-5} to ensure stable and efficient parameter updates. To prevent overfitting, dropout with a rate of 0.5 was incorporated into the encoder and decoder modules, randomly deactivating neurons during training to reduce reliance on specific features. Additionally, the loss function includes a trade-off hyperparameter α , which was set to 0.5 to balance edge and degree reconstruction tasks. The masking ratio p for the self-supervised strategy was set to 0.4. These settings were used as the default configuration for all main experiments. This includes the comparisons of different feature fusion strategies, GNN layer types, and negative sampling methods.

Our experimental development was based on Python [60], and we used the PyTorch framework to construct and train the model [61]. Lastly, we conducted the training process with $2 \times$ Nvidia 2080 Ti GPUs.

Evaluation metrics

In this study, we utilize accuracy (ACC), recall (RE), precision (PRE), specificity (SPE), F1-score (F1), Matthew's correlation coefficient (MCC), and the area under the receiver operating characteristic curve (AUC) to evaluate the performance of the presented method [62]. Additionally, the false positive rate (FPR) and false negative rate (FNR) are reported to evaluate the model's misclassification behavior across different datasets. These metrics are defined as

Table 2. Performance on the test sets of RPIs datasets

Dataset	ACC	RE	PRE	SPE	F1	MCC	AUC
RPI488	0.878	0.857	0.894	0.898	0.875	0.756	0.918
RPI369	0.873	0.887	0.863	0.859	0.875	0.747	0.813
RPI2241	0.955	0.949	0.960	0.961	0.954	0.910	0.989
RPI1807	0.985	0.983	0.986	0.986	0.985	0.970	0.990
RPI1446	0.953	0.955	0.952	0.952	0.954	0.907	0.986
RPImerged	0.933	0.947	0.921	0.919	0.934	0.866	0.973

follows:

$$\text{ACC} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (32)$$

$$\text{RE} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (33)$$

$$\text{PRE} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (34)$$

$$\text{SPE} = \frac{\text{TN}}{\text{TN} + \text{FP}} \quad (35)$$

$$\text{F1} = \frac{2 \times \text{PRE} \times \text{RE}}{\text{PRE} + \text{RE}} \quad (36)$$

$$\text{MCC} = \frac{\text{TP} \times \text{TN} - \text{FP} \times \text{FN}}{\sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}} \quad (37)$$

$$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}} \quad (38)$$

$$\text{FNR} = \frac{\text{FN}}{\text{FN} + \text{TP}} \quad (39)$$

where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively.

Results and discussion

Performance evaluation

Our model's performance across the test sets of RPI488, RPI369, RPI2241, RPI1807, RPI1446, and RPImerged is summarized in Table 2 and Supplementary Table S5. On the RPI488 dataset, the model achieved an ACC of 0.878 and an MCC score of 0.756. For RPI369, the model obtained an ACC of 0.873 and an MCC score of 0.747. The performance on the RPI2241 dataset yielded an ACC of 0.955 and an MCC score of 0.910. On the RPI1807 dataset, the model reached an ACC of 0.985 and an MCC score of 0.970. The ACC and MCC for RPI1446 were 0.953 and 0.907, respectively. Additionally, on the RPImerged dataset, the model achieved an

ACC of 0.933 and an MCC score of 0.866, demonstrating its robustness on a larger and more heterogeneous dataset.

To further evaluate the generalizability of our model, we conducted five-fold cross-validation on the RPI datasets. The average performance across the folds is summarized in Table 3 and Supplementary Table S6. On the RPI488 dataset, the model achieved an average ACC of 0.880 and an average MCC score of 0.767. For the RPI369 dataset, the average ACC was 0.811, with an MCC score of 0.630. The model achieved an ACC of 0.950 and an MCC score of 0.903 on the RPI2241 dataset. On the RPI1807 dataset, the average ACC and MCC score were 0.979 and 0.958, respectively. For the RPI1446 dataset, the model reached an average ACC of 0.910 and an MCC score of 0.821. Additionally, on the RPImerged dataset, the model achieved an average ACC of 0.924 and an MCC score of 0.849. These results confirm that the proposed method maintains consistently strong performance under cross-validation, demonstrating good generalization across multiple datasets.

Notably, the MCC scores on the RPI488 and RPI369 datasets are relatively lower compared to the other datasets. This is mainly due to two factors. First, both datasets contain a limited number of interaction pairs, leading to sparse interaction networks with reduced structural information. Second, the small number of positive samples intensifies the class imbalance, which significantly affects MCC because it is highly sensitive to false positives and false negatives. These factors jointly contribute to the lower MCC values observed on these two datasets.

Ablation experiments

Evaluation of mask ratio and alpha value settings

In this study, we evaluated the impact of the mask ratio and the hyperparameter α on the model's predictive performance. Specifically, we first conducted five-fold cross-validation on each of the benchmark datasets and calculated the average performance on the test folds. These per-dataset averages were then further averaged across all datasets to produce a single representative curve for each hyperparameter setting. The results are summarized in Supplementary Fig. S3.

As shown in Supplementary Fig. S3A, the mask ratio has a substantial impact on model performance across multiple evaluation metrics. The model achieves the best overall performance when the mask ratio is set to 0.4, particularly in terms of MCC, AUC, and ACC. This suggests that moderate masking improves the model's ability to learn informative patterns by reducing noise while preserving critical structural information. Performance generally increases as the mask ratio increases from 0.1 to 0.4, but further increases beyond 0.5 lead to a consistent decline in most metrics, indicating that excessive masking can remove important connections in the graph and impair learning. Furthermore, at a mask ratio of 0.0, the lack of a reconstruction objective causes a sharp drop in AUC and ACC, while at 1.0 the encoder cannot

Table 3. Average performance of five-fold cross-validation on multiple RPI datasets, shown as mean $\pm$ standard deviation across the five folds

Dataset	ACC	RE	PRE	SPE	F1	MCC	AUC
RPI488	0.880 $\pm$ 0.052	0.864 $\pm$ 0.095	0.897 $\pm$ 0.052	0.897 $\pm$ 0.066	0.877 $\pm$ 0.055	0.767 $\pm$ 0.100	0.923 $\pm$ 0.046
RPI369	0.811 $\pm$ 0.060	0.745 $\pm$ 0.098	0.859 $\pm$ 0.061	0.876 $\pm$ 0.058	0.795 $\pm$ 0.069	0.630 $\pm$ 0.116	0.772 $\pm$ 0.071
RPI2241	0.950 $\pm$ 0.027	0.927 $\pm$ 0.050	0.973 $\pm$ 0.008	0.974 $\pm$ 0.008	0.949 $\pm$ 0.029	0.903 $\pm$ 0.051	0.987 $\pm$ 0.009
RPI1807	0.979 $\pm$ 0.007	0.986 $\pm$ 0.009	0.973 $\pm$ 0.009	0.972 $\pm$ 0.009	0.979 $\pm$ 0.007	0.958 $\pm$ 0.014	0.986 $\pm$ 0.008
RPI1446	0.910 $\pm$ 0.029	0.884 $\pm$ 0.040	0.931 $\pm$ 0.023	0.935 $\pm$ 0.022	0.907 $\pm$ 0.030	0.821 $\pm$ 0.057	0.969 $\pm$ 0.011
RPImerged	0.924 $\pm$ 0.036	0.934 $\pm$ 0.063	0.915 $\pm$ 0.017	0.914 $\pm$ 0.013	0.924 $\pm$ 0.039	0.849 $\pm$ 0.073	0.969 $\pm$ 0.022

see any positive links and performs at near-random levels. These extreme cases suggest that mask ratios between 0.1 and 0.9 offer a balance between information availability and training difficulty, which leads to improved self-supervised learning performance.

Supplementary Fig. S3B illustrates the effect of the decoder loss weight α on performance. The model achieves peak performance when $\alpha = 0.5$, with noticeable improvements in AUC, MCC, and ACC. Values of α in the range of 0.2 to 0.6 maintain stable performance across most metrics. However, setting α to extremes (such as 0.0 or 1.0) results in performance degradation, particularly in MCC, indicating that an overly imbalanced weighting between the reconstruction loss and the main classification objective can harm the model's predictive capacity. These findings highlight the importance of tuning α within a moderate range.

Analysis of different feature fusion

To highlight the importance of feature fusion in RPI prediction, this study explored the integration of different RNA and protein features from the RPImerged dataset before inputting them into the model for prediction. Specifically, six types of features were extracted from RNA sequences: NAC, Kmer, DPCP, PseDNC, PCPseDNC, and CKSNAP. From protein sequences, five types of descriptors were extracted: AAC, PAAC, CKSAAGP, QSOOrder, and DDE, as well as ESM-2 embeddings derived from a pre-trained protein language model. Each RNA feature was individually combined with a protein feature to form 36 distinct RNA-protein feature combinations.

As presented in Table 4, among the 36 single-feature combinations, the pairing of Kmer and ESM-2 achieved the best overall performance, with an ACC of 0.883 and an MCC of 0.770. In contrast, the combination of PseDNC and ESM-2 attained the highest RE of 0.940. These results demonstrate the complementarity between different RNA and protein feature types.

Furthermore, when all RNA and protein features were integrated into a combined representation, the model achieved the best overall performance, with an ACC of 0.924 and an MCC of 0.849. This significantly outperformed any individual RNA-protein feature pair. The superior performance of the fused features confirms that feature integration can effectively capture comprehensive information from both RNA and protein sequences, thereby enhancing the predictive capability of the RPI model.

Comparison of different GNN layers

We conducted ablation experiments on the RPImerged dataset to evaluate the impact of different GNN architectures on model performance. The configurations tested included a baseline model without GNN, in which node features are transformed by fully connected layers without incorporating neighborhood information, as well as several widely used GNN layers, including Graph Convolutional Network, Graph Sample and aggreGatE, Graph Isomorphism Network (GIN), Graph Attention Network

(GAT), and their combined models [23]. As shown in the Fig. 2 and Supplementary Table S7, incorporating GNN layers consistently improved performance compared to the no-GNN baseline, indicating that neighborhood aggregation plays an important role in learning informative representations for RPI prediction.

Among all configurations, the combination of GAT and GIN achieved the best performance, with an ACC of 0.924 and an MCC of 0.849. This superior performance can be attributed to the complementary strengths of these models. In the initial phase, GAT utilizes an attention mechanism to adaptively assign different weights to each node's neighbors. This approach allows for a weighted average during the initial feature extraction phase, based on the importance of neighboring nodes, thereby identifying key nodes and edges. Subsequently, GIN further integrates these features through its powerful aggregation method, which retains more information. GAT's ability to integrate information from neighboring nodes in the initial stage is complemented by GIN's detailed fusion and processing of this information in the second stage, resulting in more refined feature representations.

Investigation of different negative sampling strategies

To investigate the influence of negative sampling strategies on model performance, we conducted a comparative evaluation of five strategies: without negative sampling, random negative sampling, file-based negative sampling, mixed negative sampling, and hard negative sampling. The experiments were performed on the RPImerged dataset, and the results are summarized in Table 5.

For the setting without negative sampling, negative pairs are derived from the initial data split and the masking mechanism in the self-supervised module, without any additional sampling strategy. In the random sampling strategy, negative RNA-protein pairs are uniformly sampled from unobserved interactions, excluding those present in the test set. File-based sampling uses a predefined set of negative pairs constructed based on prior annotations. Mixed sampling combines equal proportions of file-based and randomly selected negatives. Hard negative sampling identifies negative samples that are most similar to positive pairs in the feature space, thus posing greater difficulty for the model to distinguish.

Among the five strategies, random sampling consistently achieved the best performance across most evaluation metrics, including ACC (0.924), F1 score (0.924), and AUC (0.969). Hard negative sampling also achieved promising results, though slightly lower than random sampling. The setting without negative sampling also yielded competitive performance, with an ACC of 0.903, MCC of 0.782, and F1 score of 0.931, indicating that the model maintained strong classification capability even without negative sampling. This result suggests that the negative samples generated through the masking mechanism in the self-supervised module provided sufficient contrastive signal during training. However, the improvement in ACC from 0.903

Table 4. Performance comparison of different feature fusion on RPImerged dataset, with results shown as mean ± standard deviation across five-fold cross-validation

RNA Feature	Protein Feature	ACC	RE	PRE	SPE	F1	MCC	AUC
NAC	AAC	0.618±0.077	0.289±0.196	0.870±0.065	0.946±0.044	0.400±0.213	0.311±0.117	0.798±0.040
	CKSAAGP	0.688±0.085	0.570±0.226	0.750±0.084	0.806±0.137	0.625±0.171	0.396±0.158	0.781±0.046
	DDE	0.818±0.042	0.717±0.083	0.899±0.023	0.920±0.018	0.796±0.057	0.651±0.075	0.889±0.033
	ESM-2	0.880±0.025	0.932±0.041	0.844±0.018	0.828±0.021	0.885±0.025	0.764±0.052	0.918±0.019
	PAAC	0.694±0.074	0.472±0.163	0.844±0.037	0.916±0.026	0.592±0.159	0.431±0.127	0.846±0.024
	QSOrder	0.704±0.073	0.548±0.174	0.796±0.061	0.860±0.066	0.637±0.128	0.433±0.134	0.812±0.028
Kmer	AAC	0.694±0.053	0.472±0.101	0.846±0.056	0.916±0.026	0.602±0.091	0.432±0.100	0.827±0.054
	CKSAAGP	0.699±0.080	0.514±0.161	0.807±0.049	0.884±0.013	0.620±0.135	0.427±0.144	0.805±0.041
	DDE	0.812±0.039	0.703±0.090	0.900±0.020	0.921±0.021	0.787±0.055	0.642±0.068	0.881±0.044
	ESM-2	0.883±0.023	0.936±0.029	0.846±0.019	0.829±0.020	0.889±0.022	0.770±0.047	0.922±0.020
	PAAC	0.688±0.093	0.461±0.214	0.834±0.057	0.915±0.046	0.568±0.224	0.416±0.167	0.831±0.041
	QSOrder	0.684±0.058	0.475±0.125	0.813±0.041	0.893±0.025	0.594±0.106	0.406±0.103	0.820±0.037
DPCP	AAC	0.719±0.090	0.532±0.188	0.843±0.073	0.906±0.039	0.640±0.154	0.473±0.163	0.822±0.060
	CKSAAGP	0.698±0.059	0.543±0.108	0.786±0.072	0.852±0.052	0.639±0.088	0.417±0.116	0.769±0.059
	DDE	0.772±0.091	0.629±0.182	0.875±0.040	0.915±0.016	0.723±0.138	0.570±0.160	0.836±0.049
	ESM-2	0.843±0.044	0.870±0.089	0.825±0.016	0.817±0.015	0.846±0.049	0.690±0.091	0.899±0.024
	PAAC	0.698±0.018	0.496±0.032	0.831±0.023	0.899±0.015	0.621±0.029	0.432±0.036	0.799±0.015
	QSOrder	0.682±0.060	0.489±0.127	0.791±0.037	0.875±0.009	0.599±0.108	0.394±0.108	0.800±0.039
PseDNC	AAC	0.641±0.066	0.365±0.167	0.824±0.092	0.917±0.053	0.487±0.150	0.341±0.114	0.767±0.091
	CKSAAGP	0.689±0.087	0.565±0.224	0.751±0.077	0.813±0.116	0.623±0.181	0.396±0.161	0.788±0.055
	DDE	0.824±0.046	0.734±0.093	0.895±0.025	0.914±0.019	0.804±0.062	0.661±0.083	0.891±0.036
	ESM-2	0.880±0.024	0.940±0.032	0.840±0.021	0.821±0.026	0.887±0.023	0.767±0.049	0.920±0.019
	PAAC	0.697±0.075	0.472±0.159	0.852±0.039	0.923±0.022	0.595±0.157	0.439±0.130	0.852±0.036
	QSOrder	0.704±0.072	0.549±0.171	0.795±0.061	0.859±0.066	0.639±0.125	0.433±0.132	0.810±0.029
PCPseDNC	AAC	0.659±0.077	0.396±0.182	0.836±0.079	0.922±0.046	0.518±0.171	0.374±0.134	0.767±0.114
	CKSAAGP	0.686±0.083	0.562±0.228	0.755±0.086	0.811±0.141	0.620±0.170	0.396±0.154	0.782±0.050
	DDE	0.824±0.047	0.735±0.098	0.894±0.025	0.914±0.020	0.804±0.064	0.661±0.085	0.891±0.036
	ESM-2	0.879±0.026	0.934±0.036	0.841±0.020	0.824±0.022	0.885±0.025	0.763±0.053	0.919±0.019
	PAAC	0.699±0.075	0.489±0.170	0.840±0.036	0.910±0.032	0.604±0.160	0.438±0.127	0.848±0.031
	QSOrder	0.701±0.071	0.543±0.171	0.793±0.060	0.859±0.066	0.633±0.125	0.428±0.130	0.812±0.029
CKSNAP	AAC	0.645±0.056	0.368±0.145	0.853±0.124	0.922±0.067	0.495±0.136	0.356±0.102	0.794±0.070
	CKSAAGP	0.699±0.081	0.524±0.162	0.800±0.062	0.875±0.029	0.625±0.128	0.426±0.149	0.802±0.044
	DDE	0.810±0.034	0.695±0.073	0.902±0.020	0.924±0.018	0.783±0.047	0.638±0.059	0.882±0.040
	ESM-2	0.880±0.023	0.938±0.037	0.841±0.017	0.823±0.020	0.887±0.023	0.766±0.048	0.917±0.019
	PAAC	0.686±0.091	0.489±0.250	0.809±0.065	0.883±0.089	0.575±0.229	0.409±0.162	0.816±0.033
	QSOrder	0.684±0.055	0.468±0.110	0.821±0.039	0.900±0.019	0.592±0.099	0.408±0.099	0.819±0.037
Combined features	Combined features	0.924±0.036	0.934±0.063	0.915±0.017	0.914±0.013	0.924±0.039	0.849±0.073	0.969±0.022

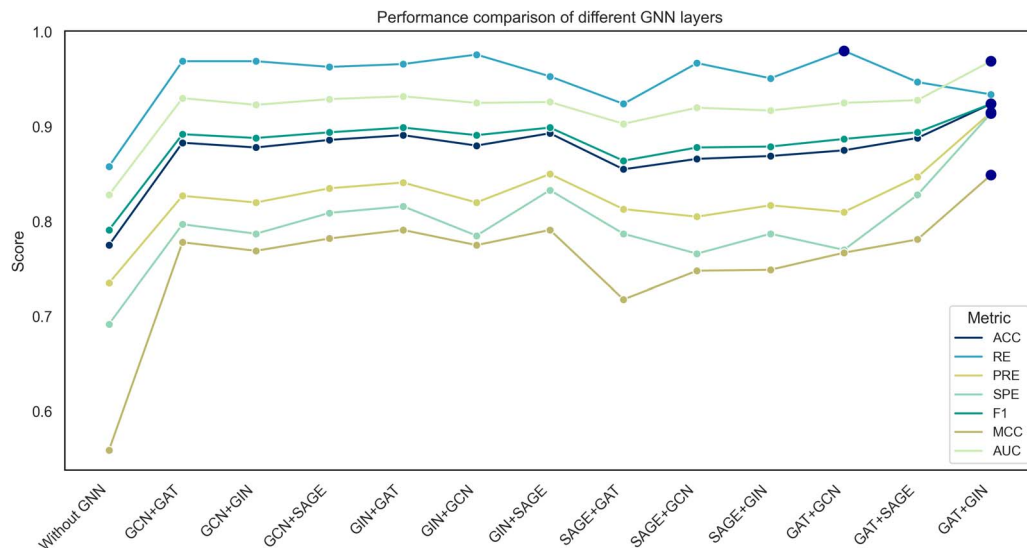


Figure 2. Performance comparison of different GNN layers. The dark blue nodes mark the highest point of each metric for different combinations of GNN layers.

Table 5. Performance evaluation of different negative sampling strategies on RPImerged dataset, with results shown as mean $\pm$ standard deviation across five-fold cross-validation

Negative sampling strategies	ACC	RE	PRE	SPE	F1	MCC	AUC
Without negative sampling	0.903 $\pm$ 0.058	0.898 $\pm$ 0.080	0.968 $\pm$ 0.008	0.918 $\pm$ 0.020	0.931 $\pm$ 0.044	0.782 $\pm$ 0.112	0.966 $\pm$ 0.020
File	0.857 $\pm$ 0.034	0.929 $\pm$ 0.042	0.813 $\pm$ 0.028	0.786 $\pm$ 0.032	0.867 $\pm$ 0.033	0.723 $\pm$ 0.070	0.910 $\pm$ 0.043
Hard	0.922 $\pm$ 0.037	0.930 $\pm$ 0.065	0.914 $\pm$ 0.014	0.913 $\pm$ 0.010	0.921 $\pm$ 0.039	0.845 $\pm$ 0.074	0.967 $\pm$ 0.024
Mixed	0.895 $\pm$ 0.027	0.934 $\pm$ 0.053	0.867 $\pm$ 0.009	0.857 $\pm$ 0.006	0.899 $\pm$ 0.029	0.794 $\pm$ 0.057	0.944 $\pm$ 0.028
Random	0.924 $\pm$ 0.036	0.934 $\pm$ 0.063	0.915 $\pm$ 0.017	0.914 $\pm$ 0.013	0.924 $\pm$ 0.039	0.849 $\pm$ 0.073	0.969 $\pm$ 0.022

Table 6. Performance evaluation of supervised and self-supervised regimes on RPImerged dataset, with results shown as mean $\pm$ standard deviation across five-fold cross-validation

Regime	ACC	RE	PRE	SPE	F1	MCC	AUC	Training time (s)
Supervised	0.768 $\pm$ 0.016	0.670 $\pm$ 0.049	0.836 $\pm$ 0.039	0.866 $\pm$ 0.045	0.742 $\pm$ 0.023	0.549 $\pm$ 0.031	0.829 $\pm$ 0.050	37.0
Self-Supervised	0.924 $\pm$ 0.036	0.934 $\pm$ 0.063	0.915 $\pm$ 0.017	0.914 $\pm$ 0.013	0.924 $\pm$ 0.039	0.849 $\pm$ 0.073	0.969 $\pm$ 0.022	38.8

to 0.924 and in MCC from 0.782 to 0.849 under random sampling highlights its additional benefits in enhancing model calibration and reducing false positives. These enhancements suggest that random sampling enables a more effective trade-off between PRE and RE, thereby contributing to improved overall robustness. In contrast, file-based and mixed sampling strategies produced relatively weaker performance across most metrics.

These results demonstrate that while the model can achieve competitive performance without negative sampling, the choice of random sampling strategy still has a measurable impact on overall model performance.

Comparison of supervised and self-supervised regimes

To highlight the advantages of our self-supervised learning framework, we conducted an experiment comparing it against a conventional supervised training regime on the RPImerged dataset. In the supervised setting, the model is trained using fixed positive and negative edge labels. In contrast, the self-supervised variant introduces two key components: negative sampling, which resamples negative pairs at each training epoch, and a masking strategy that encourages the model to learn from partially observed graph structures.

As shown in Table 6, the self-supervised approach consistently outperforms the supervised baseline across all evaluation metrics. For example, ACC improves from 0.768 to 0.924, MCC increases from 0.549 to 0.849, and AUC rises from 0.829 to 0.969. These results indicate that incorporating masking and dynamic sampling strategies helps the model leverage richer structural information and a more varied set of training examples.

It is worth noting that the improved performance of the self-supervised model does not lead to a significant increase in training cost. The average training time over 100 epochs increases slightly, from 37.0 seconds in the supervised setting to 38.8 seconds in the self-supervised setting, suggesting that the proposed approach remains computationally feasible.

Overall, these results demonstrate the effectiveness of the self-supervised learning framework and indicate its potential as a reliable alternative to conventional supervised training for RPI prediction.

Model interpretability analysis

To enhance the interpretability of our model and to better understand the contribution of different input features to prediction outcomes, we employed SHAP (Shapley Additive Explanations) to

quantify the contribution of each input feature to the model's predictions for specific RPI pairs [63]. SHAP values were computed using the Captum interpretability library [64], which supports gradient-based attribution tailored for deep learning architectures. To demonstrate the interpretability of our framework, we selected two representative RNA-protein pairs for detailed case analysis: Protein 1S1I-W with RNA 1S1I-3 (Supplementary Fig. S4) and Protein 1FFK-P with RNA 1FFK-0 (Supplementary Fig. S5).

For each case, input features were grouped into biologically meaningful categories. RNA descriptors included NAC, Kmer, DPCP, PseDNC, PCPseDNC, and CKSNAP, while protein descriptors consisted of AAC, PAAC, CKSAAGP, QSOrder, DDE, and pretrained ESM-2 embeddings. We then aggregated SHAP values within each group to evaluate their overall contribution to the prediction. To further examine the internal decision process, we ranked all features by their SHAP values and visualized the top 20 most influential ones for each pair.

In the case of Supplementary Fig. S4, the NAC feature group showed the highest average SHAP value among RNA descriptors. Within this group, the feature NAC_A exhibited the strongest individual contribution, indicating its relevance to the model's decision. On the protein side, while the group-level importance scores across categories such as DDE, ESM-2, and QSOrder were relatively close, individual features from DDE and ESM-2 were ranked highest in terms of SHAP values, suggesting their significant influence on the final output. Similarly, in Supplementary Fig. S5, the NAC group remained the most dominant contributor among RNA features, again with NAC_A ranking highest. For proteins, the most influential individual features were mainly derived from ESM embeddings and DDE descriptors, including ESM_123, ESM_48, and DDE227. These findings indicate that while group-level contributions may be balanced, certain individual descriptors carry greater weight in driving model predictions.

To further explore why models based on specific feature combinations, such as DPCP+ACC or PseDNC+AAC, perform poorly as shown in Table 4, we analyzed representative RNA-protein pairs that were misclassified under these combinations but correctly predicted when using the full set of features. Specifically, Protein 1HR0-I with RNA 1HR0-A failed to be predicted accurately when using only DPCP+ACC features, and Protein 3PYU-I with RNA 3PYU-A was misclassified under PseDNC+AAC (Supplementary Figs S6 and S7). However, when all features were combined, both predictions became correct, suggesting that additional feature types provided necessary complementary information.

SHAP analysis was performed on these two examples to understand the key factors behind the improved predictions. For both protein nodes, the ACC feature group had very low importance. In contrast, DDE and ESM-2 contributed substantially to the final predictions. For RNAs, NAC had the highest contribution, while the other five RNA feature types had relatively low importance.

A closer inspection of the top 20 individual features by SHAP value further supported these observations. For the protein nodes, most of the top contributors originated from DDE and ESM-2, with additional features from CKSAAGP and QSOOrder. On the RNA side, while NAC remained dominant, a few individual features from the DPCP group also exhibited high contributions. This indicates that although certain feature groups may show low overall importance, specific descriptors within them can still play a critical role when combined with other features.

Together, these analyses demonstrate that models using only DPCP+ACC or PseDNC+AAC are often not able to capture the complexity of RPis. In contrast, combining diverse and complementary feature groups allows the model to learn richer and more complete feature representations, which leads to more accurate predictions. These results show the importance of using multi-view feature fusion and help explain how different types of features affect model performance at both the overall and individual levels.

Analysis of Graph-encoded features for RNA–protein interactions

To analyze the graph-encoded features for RPis, we extracted RPI features from five datasets (RPI488, RPI369, RPI2241, RPI1807, and RPI1446) using our model and visualized these high-dimensional features using t-distributed Stochastic Neighbor Embedding (t-SNE) for dimensionality reduction [65]. The t-SNE visualizations reveal the distribution patterns of positive samples (RNA–protein pairs with interactions) and negative samples (RNA–protein pairs without interactions) in a two-dimensional space. As observed across all five datasets on Fig. 3, positive and negative samples are generally well-separated with only minimal overlap, indicating that the model has effectively learned the interaction features of these datasets. In the RPI488 (Fig. 3A) and RPI369 (Fig. 3B) datasets, although good separation between positive and negative samples is achieved, the sample points are relatively dispersed, which may be attributed to the smaller dataset size. In contrast, for the RPI2241 (Fig. 3C), RPI1807 (Fig. 3D), and RPI1446 (Fig. 3E) datasets, where the number of RNA–protein pairs is larger, most samples show a more clustered pattern within the same class. This suggests that our model effectively captures RPI features and can learn more robust feature representations, particularly in larger datasets. In addition, the visualization of the RPImerged dataset (Fig. 3F), which integrates all five datasets, also demonstrates a clear separation between interaction and non-interaction pairs. This highlights the model's ability to generalize its feature encoding across heterogeneous data sources and reinforces its robustness when applied to more diverse RPI scenarios. Although t-SNE does not directly evaluate the model's classification performance, it provides an intuitive understanding of the model's capability to learn discriminative feature embeddings, helping to identify performance differences across various datasets and guiding future model optimization efforts.

Comparison with other existing RPI prediction methods

To ensure a fair and rigorous comparison, we selected several state-of-the-art and widely adopted RPI prediction methods that could be consistently reproduced across all datasets. These

include RPI-CapsuleGAN [19], LPI-SKMSC [20], RPI-MDLStack [18], LPI-CNNCP [15], RPISeq-RF [14], and RPISeq-SVM [14]. All baseline models were carefully reproduced based on the authors' publicly available source code and detailed instructions in their original publications.

As shown in Table 7 and Supplementary Table S8, we observe that our proposed model (Graph-RPI) outperforms all competing methods on the RPI369, RPI1446, RPI1807, RPI2241, and RPImerged datasets. The performance advantages of Graph-RPI are especially pronounced on the larger and more complex datasets (RPI1446, RPI1807, RPI2241, and RPImerged), highlighting its robustness and scalability in modeling realistic and large-scale RPI networks.

On the RPI1446 dataset, Graph-RPI achieves an average ACC of 0.910, outperforming the baseline methods by 4 to 29 percentage points. The MCC is 0.821, which is 7 to 67 percentage points higher than those of the competing approaches. On the RPI1807 dataset, our model achieves an ACC of 0.979 and an MCC of 0.958, significantly exceeding all baselines, whose MCC values range only from 0.202 to 0.475. Similarly, on RPI2241, Graph-RPI attains an ACC of 0.950 and an MCC of 0.903, outperforming all competing methods by at least 10 percentage points in MCC. On the largest dataset, RPImerged, our model achieves an ACC of 0.924 and an MCC of 0.849, while the best-performing baseline (RPI-CapsuleGAN) only reaches 0.858 in ACC and 0.706 in MCC. Baseline methods such as RPISeqRF and RPISeqSVM exhibit a substantial performance drop on the RPImerged dataset, which can be attributed to two main factors. First, the merged dataset introduces considerable distributional heterogeneity across species, RNA categories, and sequence lengths, making it challenging for models that rely on fixed handcrafted features to generalize. Second, baseline methods typically predict each RNA–protein pair independently, overlooking the many-to-many interaction structure inherent in biological systems. In contrast, Graph-RPI models the entire dataset as a heterogeneous graph and leverages message passing to capture global interaction patterns, resulting in improved robustness and generalization.

In addition, although our performance on the RPI488 dataset was slightly inferior to RPISeqSVM, this can be attributed to the smaller number of RNA nodes in this dataset. Our method, which is based on graph networks, contrasts with previous methods that constructed balanced datasets to achieve better and more balanced results on smaller datasets. However, our approach does not require the construction of negative samples, thus aligning more closely with the real-world structure of RPI data.

To rigorously evaluate the statistical significance of the observed performance gains, we conducted paired t-tests between Graph-RPI and each baseline method across all datasets and evaluation metrics [66, 67]. For each dataset, we performed pairwise comparisons using results from five independent cross-validation folds, covering standard metrics such as ACC, F1, AUC, and MCC.

The results of the statistical analysis, as summarized in Supplementary Table S9, reveal that Graph-RPI achieves statistically significant improvements, particularly in MCC, AUC, and F1, on several datasets. For instance, on RPI2241, Graph-RPI significantly outperforms all baseline methods on nearly all metrics. Similar trends are observed on RPI1807, RPI1446, and RPImerged, where p-values are below 0.01 for multiple critical evaluation metrics, indicating that the improvements are not due to random variation.

To control for the risk of Type I errors resulting from multiple hypothesis testing, we applied the Benjamini–Hochberg correction to control the False Discovery Rate (FDR) [68, 69]. After computing raw p-values from the paired t-tests, we adjusted them using the

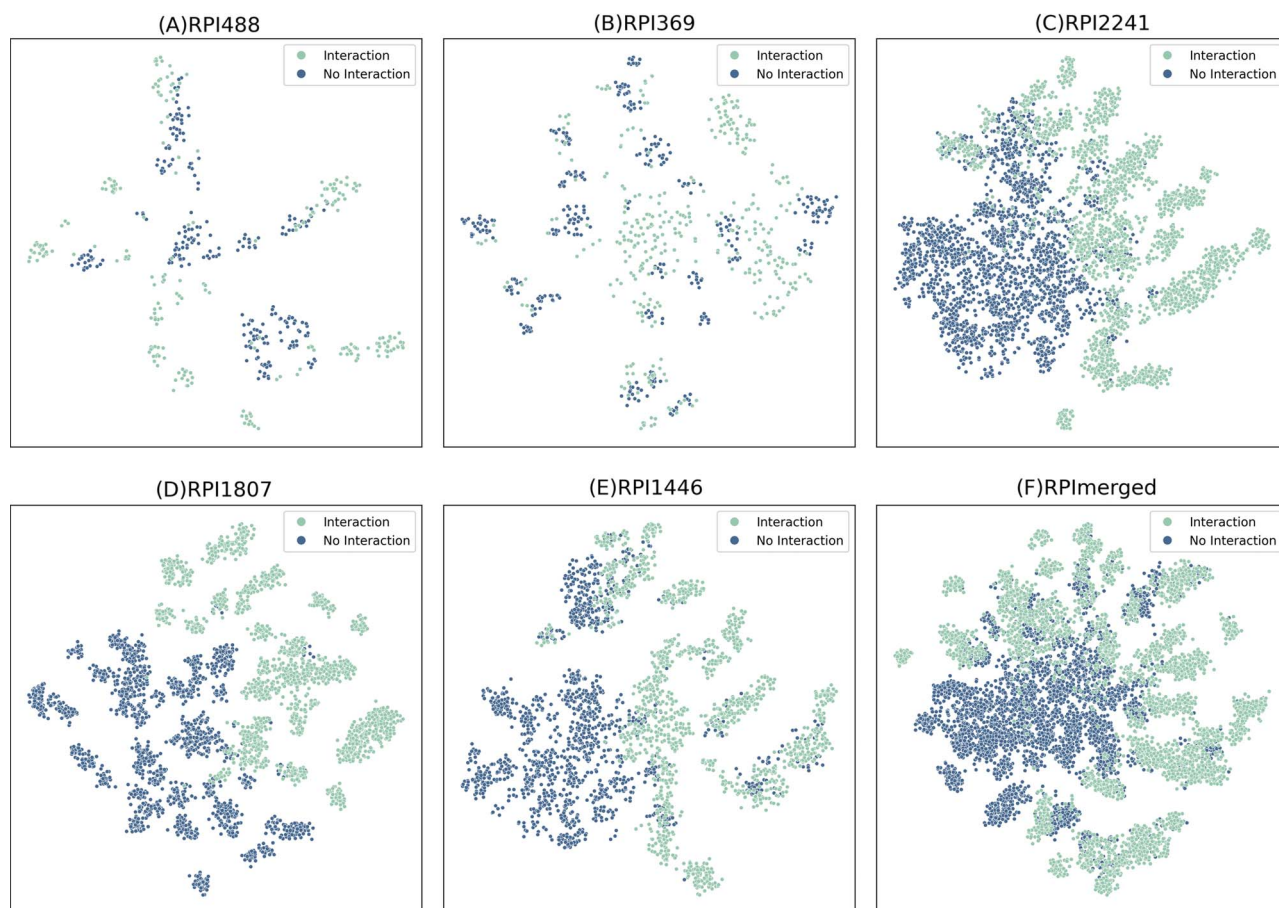


Figure 3. Visualization of encoded features for RPI prediction across six datasets: (A) RPI488, (B) RPI369, (C) RPI2241, (D) RPI1807, (E) RPI1446, and (F) RPImerged.

Benjamini–Hochberg procedure with a significance threshold of $\alpha = 0.05$. This correction ensures that the expected proportion of false positives among all rejected hypotheses remains acceptably low, thereby enhancing the reliability of our statistical conclusions.

Even after applying FDR correction, Graph-RPI retains advantages over the baseline methods across many datasets. In particular, on RPI2241, RPI1807, RPI1446, and RPImerged, a substantial number of evaluation metrics show adjusted p-values below the corrected significance threshold, confirming that the performance gains of our model are both consistent and statistically valid.

Performance evaluation on independent test sets

To verify the generalization capability of our model on multi-species datasets and to assess its ability to explore and predict unknown RPis, we used the RPI prediction model trained on RPI1807 to predict 10 093 RPI pairs from six independent test datasets. The prediction results were compared with those of RPI-CapsuleGAN [19], LPI-SKMSC [20], RPI-MDLStack [18], LPI-CNNCP [15], RPISeq-RF [14], and RPISeq-SVM [14], as detailed in Fig. 4 and Supplementary Table S10.

Our method achieved the highest performance on all datasets. Notably, our model accurately predicted all RPis in RPI_M and RPI_C, and achieved an ACC of 0.990 on RPI_H. Across the total 10 093 RPI pairs in the combined datasets, our method successfully predicted 9986 pairs, yielding an overall ACC of

0.989. This represents an improvement of 0.011 to 0.499 over other methods.

To further evaluate the model's generalization ability on more challenging cases, we computed sequence similarity matrices between each independent dataset and the RPI1807 dataset, and selected the 20% of RNA–protein pairs with the lowest similarity scores for evaluation. These results are summarized in Supplementary Table S11. The model continued to perform well on these difficult subsets, correctly predicting 1413 out of 1463 pairs in RPI_H, 105 out of 134 in RPI_S, 34 out of 36 in RPI_E, and 12 out of 13 in RPI_D. In both RPI_M and RPI_C, all selected pairs were correctly predicted.

The prediction results on the multi-species datasets demonstrate that our method possesses strong generalization ability and stable predictive performance, which contributes to advancing research in RPI prediction.

Prediction of high probability RNA–protein interaction networks

Predicting RPI networks is crucial for understanding the intricate relationships between RNA and proteins, their structural and functional roles, and the underlying mechanisms of their interactions in various biological processes. In this study, we employed our model to predict the top 30 RPI pairs with the highest interaction probabilities across six species datasets. Each dataset represents RNAs and proteins as nodes and their interactions as edges in the RPI network (Supplementary Fig. S8). In these network diagrams, green nodes represent RNAs, blue nodes

Table 7. Performance comparison with other RPI prediction methods on multiple RPI datasets, with results shown as mean $\pm$ standard deviation across five-fold cross-validation

Dataset	Method	ACC	RE	PRE	SPE	F1	MCC	AUC
RPI488	RPISeqRF	0.787 $\pm$ 0.203	0.595 $\pm$ 0.412	0.957 $\pm$ 0.046	0.979 $\pm$ 0.018	0.638 $\pm$ 0.380	0.621 $\pm$ 0.354	0.874 $\pm$ 0.158
	RPISeqSVM	0.888 $\pm$ 0.169	0.820 $\pm$ 0.319	0.910 $\pm$ 0.111	0.955 $\pm$ 0.030	0.836 $\pm$ 0.273	0.785 $\pm$ 0.319	0.936 $\pm$ 0.102
	LPI-CNNCP	0.761 $\pm$ 0.133	0.584 $\pm$ 0.276	0.889 $\pm$ 0.095	0.938 $\pm$ 0.032	0.671 $\pm$ 0.225	0.562 $\pm$ 0.241	0.883 $\pm$ 0.144
	RPI-MDLStack	0.787 $\pm$ 0.195	0.591 $\pm$ 0.392	0.965 $\pm$ 0.038	0.984 $\pm$ 0.016	0.648 $\pm$ 0.354	0.625 $\pm$ 0.338	0.873 $\pm$ 0.113
	LPI-SKMSC	0.861 $\pm$ 0.180	0.784 $\pm$ 0.311	0.873 $\pm$ 0.188	0.935 $\pm$ 0.061	0.813 $\pm$ 0.272	0.720 $\pm$ 0.370	0.897 $\pm$ 0.178
	RPI-CapsuleGAN	0.826 $\pm$ 0.155	0.697 $\pm$ 0.306	0.946 $\pm$ 0.062	0.974 $\pm$ 0.025	0.765 $\pm$ 0.261	0.689 $\pm$ 0.282	0.861 $\pm$ 0.131
	Graph-RPI	0.880 $\pm$ 0.052	0.864 $\pm$ 0.095	0.897 $\pm$ 0.052	0.897 $\pm$ 0.066	0.877 $\pm$ 0.055	0.767 $\pm$ 0.100	0.923 $\pm$ 0.046
RPI369	RPISeqRF	0.784 $\pm$ 0.173	0.620 $\pm$ 0.346	0.907 $\pm$ 0.086	0.948 $\pm$ 0.036	0.677 $\pm$ 0.311	0.601 $\pm$ 0.310	0.897 $\pm$ 0.129
	RPISeqSVM	0.806 $\pm$ 0.127	0.719 $\pm$ 0.244	0.846 $\pm$ 0.092	0.894 $\pm$ 0.020	0.761 $\pm$ 0.206	0.622 $\pm$ 0.240	0.898 $\pm$ 0.065
	LPI-CNNCP	0.774 $\pm$ 0.036	0.895 $\pm$ 0.133	0.727 $\pm$ 0.038	0.660 $\pm$ 0.078	0.794 $\pm$ 0.046	0.582 $\pm$ 0.096	0.807 $\pm$ 0.032
	RPI-MDLStack	0.663 $\pm$ 0.190	0.656 $\pm$ 0.242	0.937 $\pm$ 0.021	0.606 $\pm$ 0.266	0.744 $\pm$ 0.194	0.191 $\pm$ 0.125	0.694 $\pm$ 0.078
	LPI-SKMSC	0.741 $\pm$ 0.043	0.803 $\pm$ 0.064	0.908 $\pm$ 0.090	0.305 $\pm$ 0.262	0.846 $\pm$ 0.027	0.088 $\pm$ 0.153	0.381 $\pm$ 0.214
	RPI-CapsuleGAN	0.634 $\pm$ 0.200	0.653 $\pm$ 0.239	0.882 $\pm$ 0.064	0.385 $\pm$ 0.195	0.731 $\pm$ 0.197	0.027 $\pm$ 0.129	0.512 $\pm$ 0.144
	Graph-RPI	0.811 $\pm$ 0.060	0.745 $\pm$ 0.098	0.859 $\pm$ 0.061	0.876 $\pm$ 0.058	0.795 $\pm$ 0.069	0.630 $\pm$ 0.116	0.772 $\pm$ 0.071
RPI2241	RPISeqRF	0.691 $\pm$ 0.193	0.429 $\pm$ 0.391	0.767 $\pm$ 0.251	0.953 $\pm$ 0.031	0.480 $\pm$ 0.378	0.406 $\pm$ 0.379	0.770 $\pm$ 0.221
	RPISeqSVM	0.746 $\pm$ 0.219	0.623 $\pm$ 0.382	0.715 $\pm$ 0.294	0.869 $\pm$ 0.086	0.645 $\pm$ 0.354	0.483 $\pm$ 0.460	0.786 $\pm$ 0.232
	LPI-CNNCP	0.703 $\pm$ 0.014	0.639 $\pm$ 0.032	0.734 $\pm$ 0.025	0.767 $\pm$ 0.033	0.682 $\pm$ 0.015	0.410 $\pm$ 0.027	0.774 $\pm$ 0.009
	RPI-MDLStack	0.854 $\pm$ 0.015	0.831 $\pm$ 0.010	0.871 $\pm$ 0.025	0.876 $\pm$ 0.025	0.850 $\pm$ 0.014	0.708 $\pm$ 0.030	0.877 $\pm$ 0.010
	LPI-SKMSC	0.622 $\pm$ 0.186	0.507 $\pm$ 0.352	0.672 $\pm$ 0.276	0.548 $\pm$ 0.287	0.555 $\pm$ 0.334	0.029 $\pm$ 0.207	0.527 $\pm$ 0.240
	RPI-CapsuleGAN	0.847 $\pm$ 0.005	0.850 $\pm$ 0.012	0.845 $\pm$ 0.002	0.844 $\pm$ 0.005	0.848 $\pm$ 0.006	0.695 $\pm$ 0.009	0.871 $\pm$ 0.009
	Graph-RPI	0.950 $\pm$ 0.027	0.927 $\pm$ 0.050	0.973 $\pm$ 0.008	0.974 $\pm$ 0.008	0.949 $\pm$ 0.029	0.903 $\pm$ 0.051	0.987 $\pm$ 0.009
RPI1807	RPISeqRF	0.662 $\pm$ 0.182	0.386 $\pm$ 0.359	0.805 $\pm$ 0.228	0.960 $\pm$ 0.030	0.453 $\pm$ 0.347	0.381 $\pm$ 0.340	0.820 $\pm$ 0.210
	RPISeqSVM	0.724 $\pm$ 0.198	0.571 $\pm$ 0.351	0.772 $\pm$ 0.273	0.894 $\pm$ 0.086	0.624 $\pm$ 0.318	0.472 $\pm$ 0.403	0.822 $\pm$ 0.212
	LPI-CNNCP	0.557 $\pm$ 0.057	0.250 $\pm$ 0.105	0.802 $\pm$ 0.135	0.945 $\pm$ 0.011	0.374 $\pm$ 0.150	0.243 $\pm$ 0.129	0.866 $\pm$ 0.052
	RPI-MDLStack	0.653 $\pm$ 0.162	0.387 $\pm$ 0.284	0.948 $\pm$ 0.046	0.988 $\pm$ 0.009	0.499 $\pm$ 0.278	0.437 $\pm$ 0.252	0.879 $\pm$ 0.085
	LPI-SKMSC	0.665 $\pm$ 0.171	0.544 $\pm$ 0.299	0.747 $\pm$ 0.253	0.671 $\pm$ 0.294	0.613 $\pm$ 0.291	0.202 $\pm$ 0.313	0.650 $\pm$ 0.266
	RPI-CapsuleGAN	0.677 $\pm$ 0.168	0.438 $\pm$ 0.298	0.943 $\pm$ 0.035	0.980 $\pm$ 0.005	0.548 $\pm$ 0.267	0.475 $\pm$ 0.264	0.698 $\pm$ 0.149
	Graph-RPI	0.979 $\pm$ 0.007	0.986 $\pm$ 0.009	0.973 $\pm$ 0.009	0.972 $\pm$ 0.009	0.979 $\pm$ 0.007	0.958 $\pm$ 0.014	0.986 $\pm$ 0.008
RPI1446	RPISeqRF	0.636 $\pm$ 0.173	0.334 $\pm$ 0.341	0.749 $\pm$ 0.259	0.954 $\pm$ 0.029	0.402 $\pm$ 0.334	0.318 $\pm$ 0.337	0.778 $\pm$ 0.211
	RPISeqSVM	0.653 $\pm$ 0.227	0.472 $\pm$ 0.371	0.654 $\pm$ 0.340	0.847 $\pm$ 0.121	0.521 $\pm$ 0.351	0.313 $\pm$ 0.481	0.702 $\pm$ 0.308
	LPI-CNNCP	0.627 $\pm$ 0.050	0.601 $\pm$ 0.272	0.684 $\pm$ 0.141	0.655 $\pm$ 0.205	0.560 $\pm$ 0.209	0.287 $\pm$ 0.062	0.720 $\pm$ 0.017
	RPI-MDLStack	0.874 $\pm$ 0.006	0.889 $\pm$ 0.015	0.855 $\pm$ 0.015	0.860 $\pm$ 0.019	0.871 $\pm$ 0.005	0.749 $\pm$ 0.011	0.912 $\pm$ 0.004
	LPI-SKMSC	0.624 $\pm$ 0.166	0.454 $\pm$ 0.303	0.688 $\pm$ 0.246	0.708 $\pm$ 0.263	0.528 $\pm$ 0.293	0.153 $\pm$ 0.286	0.623 $\pm$ 0.247
	RPI-CapsuleGAN	0.842 $\pm$ 0.009	0.858 $\pm$ 0.017	0.822 $\pm$ 0.008	0.828 $\pm$ 0.010	0.840 $\pm$ 0.010	0.685 $\pm$ 0.018	0.866 $\pm$ 0.012
	Graph-RPI	0.910 $\pm$ 0.029	0.884 $\pm$ 0.040	0.931 $\pm$ 0.023	0.935 $\pm$ 0.022	0.907 $\pm$ 0.030	0.821 $\pm$ 0.057	0.969 $\pm$ 0.011
RPImerged	RPISeqRF	0.554 $\pm$ 0.008	0.039 $\pm$ 0.022	0.236 $\pm$ 0.120	0.920 $\pm$ 0.012	0.066 $\pm$ 0.037	-0.088 $\pm$ 0.048	0.300 $\pm$ 0.060
	RPISeqSVM	0.505 $\pm$ 0.016	0.042 $\pm$ 0.033	0.137 $\pm$ 0.093	0.833 $\pm$ 0.028	0.064 $\pm$ 0.049	-0.196 $\pm$ 0.054	0.229 $\pm$ 0.064
	LPI-CNNCP	0.693 $\pm$ 0.016	0.531 $\pm$ 0.079	0.676 $\pm$ 0.046	0.809 $\pm$ 0.066	0.589 $\pm$ 0.034	0.361 $\pm$ 0.027	0.766 $\pm$ 0.007
	RPI-MDLStack	0.628 $\pm$ 0.049	0.192 $\pm$ 0.121	0.652 $\pm$ 0.102	0.940 $\pm$ 0.005	0.287 $\pm$ 0.145	0.187 $\pm$ 0.131	0.810 $\pm$ 0.017
	LPI-SKMSC	0.518 $\pm$ 0.012	0.146 $\pm$ 0.012	0.339 $\pm$ 0.032	0.792 $\pm$ 0.009	0.203 $\pm$ 0.016	-0.080 $\pm$ 0.023	0.425 $\pm$ 0.059
	RPI-CapsuleGAN	0.858 $\pm$ 0.003	0.797 $\pm$ 0.009	0.853 $\pm$ 0.011	0.901 $\pm$ 0.010	0.824 $\pm$ 0.003	0.706 $\pm$ 0.006	0.864 $\pm$ 0.009
	Graph-RPI	0.924 $\pm$ 0.036	0.934 $\pm$ 0.063	0.915 $\pm$ 0.017	0.914 $\pm$ 0.013	0.924 $\pm$ 0.039	0.849 $\pm$ 0.073	0.969 $\pm$ 0.022

represent proteins, and edges illustrate predicted interactions, providing valuable insights for interpretability analysis.

As depicted in [Supplementary Fig. S8A](#), a total of 30 high-probability RPI pairs were predicted in the *H. sapiens* dataset. These include interactions between proteins such as P42574 (Caspase-3) and P61964 (a WD40 repeat protein), and RNAs like AC006369.2 and TRIM36-IT1. Each predicted pair suggests a potential regulatory mechanism with implications for cellular function and disease.

To further evaluate the biological plausibility of these predictions, we selected the aforementioned RNA-protein pairs and employed AlphaFold3 to predict their structural conformations [70]. Additionally, the experimentally validated interaction between CDKN2B-AS1 and Q15022 was used as a positive control for comparison. As shown in [Supplementary Fig. S9](#), the predicted complexes exhibited consistent interaction patterns, with specific amino acid residues forming hydrogen bonds with corresponding RNA nucleotides [71]. The binding interfaces of the predicted pairs

closely resembled those of the validated complex, indicating both structural feasibility and potential biological relevance.

Building upon these structural insights, we further explored the biological context of the predicted RNA-protein pairs. For instance, P42574 (Caspase-3) is a key cysteine protease involved in the execution phase of apoptosis and has been linked to immunological infertility [72]. In terms of RNA, AC006369.2 is one of 16 differentially expressed ferroptosis-related lncRNAs associated with skin squamous cell carcinoma. These lncRNAs play a role in regulating the tumor immune microenvironment, potentially influencing gene expression related to ferroptosis and immune checkpoints [73]. The interaction between P42574 and AC006369.2 might suggest a crosstalk between apoptosis and ferroptosis pathways within certain tumor microenvironments, indicating that AC006369.2 may modulate Caspase-3 expression or activity through its RNA-mediated mechanisms.

Similarly, P61964, a member of the WD40 repeat protein family, is known to act as a scaffold for protein-protein interactions and

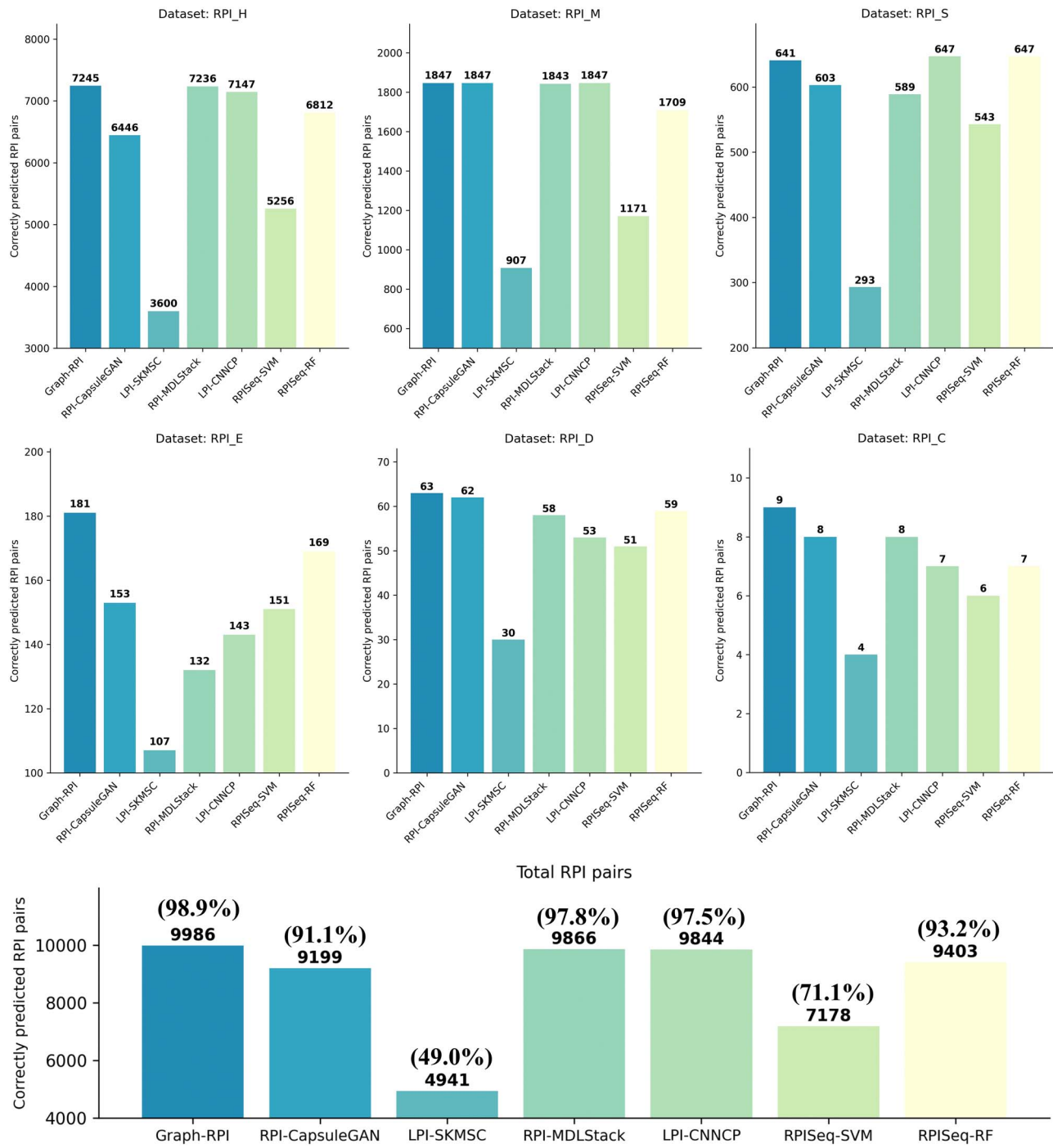


Figure 4. Performance comparison with other existing RPI prediction methods on six species datasets.

is implicated in various cellular pathways, including transcription regulation and the ubiquitin-proteasome system. WD40 repeat proteins are associated with several diseases, such as cancer, neurological disorders, metabolic syndromes, viral infections, and inflammatory conditions [74]. In parallel, TRIM36-IT1, a long non-coding RNA related to clear cell renal cell carcinoma (ccRCC), has been identified as an independent prognostic factor influencing overall survival in ccRCC patients. TRIM36-IT1 is involved in the competing endogenous RNA network, modulating specific mRNA expression by binding to microRNAs (e.g. hsa-mir-21) [75]. This regulation might influence protein interactions indirectly, thereby contributing to cancer progression.

These interactions, predicted by our model, illustrate the complex interplay between RNAs and proteins in *H. sapiens*, highlighting how specific RNA molecules and proteins may influence each other's function and regulation. By mapping these predicted interactions, we can better understand the underlying molecular mechanisms that govern cellular processes. Extending this analysis to RPI networks in other species datasets (Supplementary Fig. S8B-F) could provide further valuable insights into conserved or species-specific cellular regulatory mechanisms. Such comparative analyses could reveal fundamental principles of RPIs, help identify potential targets for therapeutic interventions, and improve our understanding of the evolutionary dynamics of these

interactions across different organisms. Additionally, the results of these high-probability RPI predictions are provided in Tables S12-S17 for further reference and detailed analysis.

Conclusion

In this study, we addressed key challenges in RPI prediction by developing a self-supervised graph-based framework that effectively integrates RNA and protein sequence information into a unified interaction graph. Traditional RPI prediction methods often suffer from limitations such as insufficient feature representation, high model complexity, and poor generalization. Our proposed model leverages GNNs to learn from the topological structure of RPI networks, where each node is iteratively updated based on its neighboring nodes, rather than treating each interaction in isolation. This design enables the model to capture global patterns in the interaction network, leading to improved predictive performance across multiple benchmark datasets.

We evaluated our model on five widely used benchmark datasets (RPI488, RPI369, RPI2241, RPI1807, and RPI1446), which are commonly adopted in the field due to their reliable annotations and well-established standards for model comparison. To further assess the generalization capability of our method, we conducted prediction experiments on independent test datasets derived from six species, including *H. sapiens*, *M. musculus*, *S. cerevisiae*, *E. coli*, *D. melanogaster*, and *C. elegans*. These datasets were collected from public resources such as NPInter and contain experimentally validated interactions that were not involved in training. The results demonstrate strong generalization performance across species and unseen data.

We also acknowledge several limitations in the current study. First, although the independent test datasets contain experimentally supported interactions, they are still curated from public databases and may not fully capture the complexity of real biological or clinical settings. Due to practical constraints related to experimental conditions and time, we have not conducted large-scale biological or clinical validation. As part of future work, we plan to collaborate with experimental or clinical teams to collect newly generated RPI data under more realistic biological conditions, which will be used to further evaluate and improve our framework.

Second, our model does not explicitly incorporate context-specific information such as cell type or tissue origin [76, 77], as the current benchmark datasets lack such annotations. However, we recognize that RPIs are often highly dependent on biological context. In future research, we aim to integrate cell- or tissue-specific datasets generated through high-throughput technologies such as CLIP-seq, RNA-seq, and proteomic profiling [78, 79]. These data will allow us to build stratified RPI networks and evaluate model performance in different biological environments.

Finally, although the current study does not directly link predicted RPIs to clinical outcomes, we recognize the importance of exploring the clinical relevance of RPI networks. As a future extension, we plan to incorporate publicly available resources such as TCGA [80], ClinVar [81], or disease-specific expression profiles to investigate whether predicted interactions correlate with disease states or other clinically meaningful features. This will support the translational potential of our framework in areas such as biomarker discovery and therapeutic target identification.

In summary, our method provides an effective and scalable solution for RPI prediction, with demonstrated performance on both benchmark and independent datasets. By extending the model to integrate biological context and clinically relevant data

in future work, we believe this framework holds promise for advancing RPI-related research and its applications in disease understanding and drug development.

Key Points

- In this study, We proposed a novel RNA–protein interaction (RPI) prediction framework using graph autoencoder, which integrates RNA and protein sequence features into a unified graph structure.
- The model employs self-supervised learning strategies, including masking mechanism and negative sampling, to enhance feature representation and improve prediction robustness without relying on explicit negative sample construction.
- Extensive experiments demonstrate that our model achieves strong predictive performance across multiple datasets, achieving results comparable to or exceeding existing RPI prediction methods
- Our approach shows strong generalization capability across multi-species datasets, suggesting its potential for broad applications in disease-related RPI research and therapeutic target identification.

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Author contributions

J.H.G. and Y.-C.C. presented the idea. J.H.G. implemented the framework. J.H.G., P.L.X. and Z.H.Z. collected the data. J.H.G. and D.M. analyzed the predictive results. Y.-C.C., J.W.W., T.-Y.L., and L.T.Y. provided expert guidance for the experiments. Y.-C.C., J.W.W., and T.-Y.L. supervised the research project.

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Data availability

The full experimental code and datasets are available at <https://github.com/GGCL7/Graph-RPI-Experiments>. The standalone software for single or multiple RNA–protein pair prediction is available at <https://github.com/GGCL7/Graph-RPI>.

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